

Vuka: DSAC Public Entities Reporting — PRD

GovTech 2026 Hackathon · 17 September 2026

1. Problem and context

The Department of Sport, Arts and Culture funds 28 bodies through Vote 37, plus a separate set of non-profit institutions. Three counts circulate and all three are correct about different things: the challenge statement says 26 public entities and six NPOs, National Treasury's Estimates of National Expenditure lists 28 bodies receiving entity transfers, and the Auditor-General audited 29 in this portfolio, being those 28 plus the department itself. This document uses 28, because that is what the published transfer table contains and what the system is seeded with. Each is obliged to report back to DSAC on two things: performance against agreed targets, and accountability for the funds received.

There is no digital system for this. Reporting happens by email, in documents written to whatever format each entity chooses.

Three consequences follow, all stated in the challenge brief.

Entities miss deadlines. Manual reporting over email has no enforcement, no visibility of what is outstanding, and no early signal that a submission is at risk.

DSAC cannot analyse what arrives. The reports come in inconsistent formats and the review process is entirely manual. Comparing one entity against another, or this year against last, means a person reading documents side by side. Management information is not available at the click of a button; it is not available at all without weeks of work.

The public cannot see what the money produced. The brief makes this point explicitly: the absence of a system to track allocation and usage of funds, and reporting on service delivery obligations to citizens, affects public sentiment around the services these institutions render.

That third consequence is the one most likely to be treated as background colour by other teams. We treat it as the centre of the problem. The reporting failure is not merely an administrative inconvenience inside a department. It is a break in the line between public money going out and a citizen being able to see what it bought.

2. Product concept: the standard, and the system that runs on it

Working name: Vuka (to wake, to rise). Placeholder.

One line. Government already defines what public entities must report. Nobody has made it machine-readable. Vuka does, and everything else follows from that.

The reframe

The reason DSAC cannot analyse what arrives is not that it lacks a portal. It is that 28 bodies send 28 differently-shaped documents. A portal that accepts free-form documents inherits exactly the same problem behind a nicer login page, which is how most reporting systems fail.

So the product has two halves.

The standard. A machine-readable schema for quarterly and annual performance reporting, derived from the structure entities already use in their Annual Performance Plans: outcome, output, indicator, baseline, annual target, quarterly target, actual, variance, spend. Published as a template entities fill in, and as a schema the system validates against.

The system. Everything the brief asks for, built on that spine: submission, repository, workspaces, analytics, early warning.

The standard is the part that is missing in the real world, and it is also the engineering decision that makes the rest of this buildable. A defined template is what makes parsing reliable, the data model obviously correct, and a live demo something other than a gamble.

What already exists, and where we stand

This section was added after checking the assumptions in this document against the actual landscape, and it is the most important correction in the file.

DPME already runs a quarterly performance reporting system. It is called eQPRS, it is live at eqprs.dpme.gov.za, and it does indicator and target configuration, actuals capture against those targets, an approval workflow and dashboards. DPME has been running onboarding sessions for national public entities. A successor, the Integrated Reporting System, is planned, though DPME's own annual report records that the related monitoring system was not developed due to procurement delays.

Any pitch that opens with "we built a way to capture performance data" loses in front of anyone who knows this. So we do not make that claim.

What eQPRS does not do is hold the evidence. It stores numbers. It does not store the document behind each number, the cell that number came from, or the named person who stands behind it. That is the actual failure mode in this portfolio: when the Auditor-General qualifies an opinion on performance information, it is rarely because a figure was invented, it is because nothing could be produced to support it.

And there is a genuinely unserved gap. Every DSAC body is a PFMA Schedule 3A entity, and a Schedule 3A entity reports performance to its **executive authority** under Treasury Regulation 30.2.1, which prescribes no deadline and names no system. The widely quoted 30-day rule is Treasury Regulation 26.1.1 and it covers revenue and expenditure only. eQPRS was built around departments. Nobody built for this line.

So the position is: eQPRS is where the numbers go, Vuka is where they can be defended. We do not compete on capture, we export to it. §5.2 covers the export.

The accountability chain

Once reporting is structured, four links connect that today do not:

Allocation. What each entity received this financial year, against which programme.

Targets. What it committed to deliver for that money, from its APP.

Evidence. What it actually reported, quarter by quarter, with the source behind every figure.

Outcome. What a citizen can be shown: allocation, commitment, delivery, and how recently the entity reported.

When the chain is intact, things become possible that are impossible today. Unit cost can be compared against what was planned. Under-delivery becomes visible mid-year rather than at year end. A parliamentary question can be answered in minutes.

What this is not

Not a new reporting burden. The template digitises what entities already produce. It does not add a sixth report to the five they already file.

Not a document management system. Documents are an input, not the product. If the value were storing PDFs, SharePoint exists.

Not an audit or assurance tool. We surface signals and route them to a human reviewer. Certifying performance information remains the Auditor-General's role.

Not an autonomous decision-maker. Every machine-derived figure and every risk score is a proposal shown with its reasoning, confirmed by a named human before it counts.

3. Users

Primary: the entity reporter

Finance or strategy staff at one of the 28 funded bodies. The two groups differ sharply and the product must serve both.

A large entity has a finance team, a planning unit, reliable connectivity and existing internal reporting rhythms. Its problem is duplication: it produces substantially the same information for DSAC, National Treasury, the Auditor-General and its own board, in four different shapes.

An NPO may be two or three people. Its problem is capacity: reporting competes directly with delivering the service. Connectivity may be a phone on mobile data. For this user, any system that demands an hour at a desktop portal will be used late or not at all.

What both need: to know exactly what is outstanding, to submit without rekeying work they have already done, and to know where they stand before the department tells them.

Primary: the DSAC oversight official

Sits in the department, responsible for monitoring performance across all 28 funded bodies. Today they read documents and build spreadsheets by hand.

What they need: to know which entity needs attention this week and why, to see any figure's source without asking for it, to compare entities on a like-for-like basis, and to produce management information on demand rather than on request.

Secondary: the DSAC executive

The DG, deputy DGs, and anyone who must answer to Parliament, the media or the Minister. Consumes summaries, not screens.

What they need: a short, current, defensible picture of portfolio performance, and the ability to trace any statement in it back to a source.

Secondary: the citizen

A member of the public, a journalist, a researcher or a civil-society organisation. No account, no training, often on a low-end phone and limited data.

What they need: to see what a given entity received, what it promised, what it delivered, and when it last reported, in plain language.

4. Scope: the brief's mandatory requirements

These five come directly from the challenge statement. They are worth 20 percent of the score on their own (Relevance) and underpin Technical Feasibility (20 percent). They are the floor. Everything in section 5 sits on top of them and scores nothing if the floor is missing.

a. Analytics module. A dashboard per public entity showing trends and patterns, including audit findings and year-on-year comparisons. Annual targets shown by state: in progress, not started, deadline missed. Statistics on staff demographics and jobs created per entity.

b. Early warning. Risk-based alerts drawing on reporting and historical information to identify potential delay or non-performance before deadlines are missed. Notifications on regulatory due dates and closing dates, with countdowns at 30 days, 15 days, and hourly as the deadline approaches.

c. Document repository. Storage for strategic plans, Annual Performance Plans, operational plans, annual reports, quarterly reports and financials, uploaded by entities and reviewable by DSAC.

d. Workspaces for public entities. Per-entity workspace with: integration with Microsoft technologies; version control triggered on save or upload; comments visible on screen in real time; task assignment internally and externally, up and down the operations process; approval confirmation on receipt of upload; and full functionality on mobile devices.

e. Security and privacy. Demonstrated alignment with South African cybersecurity and privacy principles. Covered in section 9.

Known scope risks

Two items in (d) are disproportionately expensive relative to their weight in the marking.

Microsoft integration. Full Microsoft 365 integration (SharePoint or Graph document sync, Teams presence) is not achievable in the build window, and pretending otherwise fails the first question.

The position, in three layers, is set out in full in §8 under *Microsoft technologies*. In short: Office document handling is **built and is the spine of the product**, because the reporting template is a real `.xlsx` workbook that the system generates and parses, and submissions export back to `.xlsx`. Entra ID single sign-on is **configuration rather than code**, because Firebase Authentication federates to Entra as an identity provider and our backend only ever reads two claims off a verified token. SharePoint and Teams are **roadmap, described as roadmap**, with a named interface where the swap happens.

The brief's judging criteria explicitly allow partial implementation where the approach is credible. The risk to accept consciously: if a judge treats "integration with Microsoft technologies" as a hard gate rather than a graded item, layer one alone has to carry it. It is a defensible thing to carry, because meeting entities in the tool they already use is a more substantive integration than a Teams tab.

Real-time comments. Live collaborative commenting requires a websocket layer that is custom work in this stack. Position: build threaded comments anchored to specific targets and documents, with near-real-time refresh. Comment-per-target is arguably more useful to a reviewer than a live cursor anyway, since it routes feedback to the thing being disputed.

5. Differentiating features

Six features beyond the floor. Each maps to a scoring criterion other than relevance, and each has been stress-tested against the question a hostile judge would actually ask.

5.1 Unit cost: planned versus actual

Every dashboard at this hackathon will show targets in red, amber and green. Ours leads with a different number: what each outcome actually cost, against what the entity said it would cost.

The APP already contains the planned figure — an entity commits to a target and to an allocation behind it. Dividing gives planned cost per unit of output. The quarterly report gives actual delivery and actual spend. The gap between planned and actual unit cost is the most useful number in the system, and nobody computes it today.

Three comparisons, in descending order of defensibility:

1. **Entity against its own plan.** Planned rand per unit versus actual. Unimpeachable, because it is the entity's own commitment.

2. **Entity against its own history.** Unit cost this year versus last. Also clean.
3. **Entity against sector peers.** Only within like-for-like groupings — museum against museum, sports body against sports body — and labelled as indicative.

What we deliberately do not do: compare cost per job across a ballet company and a sports federation. Outcomes are not commensurable across different mandates, and anyone in that room who knows public finance will say so. The interface enforces the grouping rather than offering a comparison that cannot be defended.

This is what expenditure reviews do by hand, months after the fact. We do it continuously, from data the entity has already submitted.

5.2 Report once, satisfy everyone

Entities report substantially the same underlying facts to DSAC, National Treasury, the Auditor-General and Parliament, in four different formats. That duplication is a direct cause of the missed deadlines the brief complains about.

Vuka models the facts once — target, actual, variance, spend, evidence — and generates the different output shapes from that single source. Submit to DSAC, export the Treasury view.

This is the requirement behind the requirement. It is also the feature most likely to make a real entity CFO in the room lean forward.

5.3 The template is the interface

Entities already write their quarterly reports. Asking them to rekey that into web forms is how reporting systems die. But asking a machine to read arbitrary prose is how demos die.

So we define the middle: a standard reporting template — a spreadsheet, one row per indicator, columns matching the schema. Entities fill in what they already know. The system reads it, validates it against that entity's registered targets, and shows the parsed result back for confirmation.

This is the right call rather than a compromise, for three reasons:

- **It works.** Parsing a defined template is reliable engineering. Parsing an arbitrary APP is a research project, and a research project cannot be demoed on a deadline.
- **It is the actual product insight.** The missing piece in the real world is the standard. The template is that standard in a form an NPO with three staff can use.
- **It degrades honestly.** Where a narrative document is uploaded instead, we extract what we can and flag the rest for manual entry, rather than claiming to read everything.

Nothing enters the record until a named human confirms it. Parsed values land in their own table with source location and a confidence flag. The reporter corrects what is wrong and confirms. A reviewer can open the source cell behind any figure in the system.

This converts requirement (c), the document repository, from a filing cabinet into the system's actual input mechanism.

5.4 Behavioural risk scoring

The brief asks for countdowns at 30 days, 15 days and hourly. Everyone will build that. It is a calendar, not intelligence.

On top of it, a risk score per entity derived from behaviour and history:

- Submission lateness across prior reporting periods
- Evidence gaps: targets with no supporting evidence attached, weighted by how far into the year we are
- Spend-to-delivery divergence: funds substantially drawn against targets substantially not started
- Prior audit findings, and whether they were repeat findings
- Revision churn: targets restated or reworded mid-year

The score is always shown with its contributing signals. The interface never displays a bare number. It displays: *high risk, because late in three of the last four quarters, 60 percent of the year elapsed with 20 percent of targets started, and 70 percent of allocation drawn.* A reviewer can disagree with any individual signal.

This directly addresses criterion 5's requirement for transparency and explainability, and criterion 6's requirement for human oversight.

5.5 Low-bandwidth submission

The six NPOs are small and under-resourced. A desktop portal is a barrier for exactly the bodies least able to absorb one.

What ships first: a sub-100KB mobile submission view. One indicator per screen, large touch targets, plain language, works on a low-end phone over a weak connection, and keeps the draft locally if the connection drops mid-submission. Buildable with certainty, and a substantive answer to criterion 4 on its own.

What ships second, only if the demo thread is already solid: a lightweight submission channel over WhatsApp, where an NPO sends the template or answers one prompt per indicator, landing in the same data model through the same service layer. Verified WhatsApp Business access takes weeks of Meta review, so a sandbox number is the only honest path inside a 48-hour window, and the pitch says exactly that rather than implying production readiness.

The order matters. The inclusivity claim has to rest on something that certainly works, with the striking channel on top of it rather than underneath it.

5.6 Publish-ready citizen view

A citizen-facing page per entity: allocation received, targets committed, delivery reported, date last reported. Plain language, minimal payload, no login.

Under DSAC's control, not ours. Nothing reaches the public surface that has not been through departmental review and an explicit publication decision. Whether and when to publish entity performance is a governance call belonging to the department and the Minister, not to a product. The system makes publication a switch rather than a project.

That distinction matters in the room. A team that publishes underperformance by default has not understood how government works. A team that hands the department a publish-ready view has.

This closes the loop on the brief's own statement about public sentiment, and it is the clearest expression of the citizen-outcome framing in criterion 7.

6. User journeys

6.1 Entity reporter: submitting a quarterly report

Thandi handles finance at a small arts NPO.

1. **Notified, usefully.** 30 days before the quarterly deadline she gets a notification. It does not just say a report is due; it lists which targets currently have no evidence attached. At 15 days it repeats with what is still outstanding. In the final hours it escalates.
2. **Submits what she already wrote.** She uploads her quarterly report document. The system extracts targets, actuals, variances and spend, and shows the structured result back to her.
3. **Confirms or corrects.** She fixes the one line the extraction misread and confirms. Nothing counts until she does.
4. **Or submits from a phone.** Low on data, she opens the mobile view instead and answers one short prompt per target, one indicator per screen. Same destination, same data model, and the step index sits in the URL so a dropped connection loses nothing.
5. **Sees herself as the department sees her.** Her own dashboard shows targets by state, her submission history, and her current risk score with its reasons. No surprises when DSAC calls.
6. **Receives feedback in place.** If the reviewer disputes a figure, the comment is attached to that target, not buried in an email thread.

6.2 DSAC oversight official: the Monday review

Sipho monitors all 28 funded bodies.

1. **One screen, ranked by risk.** Entities ordered by risk score, not alphabetically. The entity whose score moved this week is at the top, with the reasons beside it.
2. **Drills into the entity.** Trend over prior years, audit findings history, targets by state, staff demographics, jobs created.
3. **Checks the source.** Every figure carries a link to the document and location it was extracted from. Nothing is taken on faith.
4. **Compares against peers.** Cost per job created here versus the portfolio average. The conversation he needs to have with the board, which today he cannot have.
5. **Reviews and responds.** Approves the submission, or returns it with a comment anchored to the specific target in dispute.
6. **Generates, rather than rebuilds.** Needs the Treasury or AG cut of the same information: generates it.

6.3 Executive: preparing for a question

The DG has a parliamentary question on an entity's performance. She opens the portfolio brief: a plain-language summary of current state across the portfolio, every claim linked to the submission and document behind it. Two minutes, defensible.

6.4 Citizen: checking an entity

A journalist opens the public page for an entity. Allocation, commitments, delivery, date last reported. Plain language, no login, loads on a low-end phone.

6.5 Demo script (90 seconds)

The demo is journey 1 flowing into 2 flowing into 4, in one unbroken thread.

Time	On screen	Said
0:00	Low-bandwidth mobile view, NPO submits one indicator per screen	"An NPO with three staff and no finance system submits from a phone, on a page under 5KB."
0:15	Extraction result, targets and actuals structured, human confirms	"They upload what they already wrote. We structure it. A human confirms before anything counts."
0:35	DSAC dashboard updates live, risk score moves	"The department sees it immediately."
0:50	Risk score expanded, contributing signals visible	"And it tells you why. Late three quarters running, evidence missing, spend ahead of delivery. Not a black box."
1:10	Rand per outcome comparison across entities	"Which lets you ask the real question. Not did they report. Was the money worth it."
1:25	Public entity page	"And a citizen can see what their money bought."

Closing line, delivered separately: standard Java, Spring Boot and PostgreSQL, hosted in Johannesburg, every figure and every document in an ordinary Postgres database the department can take in-house tomorrow. No proprietary runtime, no licence per seat. And it exports straight into the system DPME already runs, so this sits alongside government's reporting rather than competing with it.

7. Data model

The model comes first: JPA entity classes define the schema, Flyway migrations match them, and Hibernate runs with `ddl-auto: validate` so the application refuses to start if the two ever drift.

Getting this model right on the first pass is most of the build.

Entity	Purpose	Key fields
PublicEntity	A funded body (26 PEs, 6 NPOs)	Name, type (PE/NPO), mandate, sector, contact person, size band
FinancialYear	Reporting year	Label (2026/27), start, end
Allocation	Money committed to an entity for a year	PublicEntity, FinancialYear, programme, amount, date approved
Target	A commitment from the APP	PublicEntity, FinancialYear, description, KPI, baseline, annual target, unit of measure, quarterly breakdown, linked Allocation
ReportingPeriod	Q1-Q4 plus annual	FinancialYear, label, period start/end, submission due date, regulatory deadline
Submission	One entity's report for one period	PublicEntity, ReportingPeriod, status (draft/submitted/under review/returned/approved), submitted date, submitted by, channel (web/mobile/WhatsApp)
TargetResult	Actual against one target for one period	Target, Submission, actual value, variance, variance explanation, spend to date, status (not started/in progress/achieved/missed)
DocumentRecord	An uploaded file	PublicEntity, document type (APP/strategic plan/quarterly/annual/financials), Submission, file, version, uploaded by, uploaded date, approval status
ExtractionResult	What was read out of a document	DocumentRecord, extracted field, extracted value, source page/location, confidence, confirmed by, confirmed date
RiskSignal	One contributing factor to a score	PublicEntity, signal type, value, weight, period observed, description
RiskScore	Computed score at a point in time	PublicEntity, ReportingPeriod, score, band, computed date, contributing RiskSignals
AuditFinding	Historical AG findings	PublicEntity, FinancialYear, finding type, description, repeat finding flag, resolution status

Entity	Purpose	Key fields
WorkforceStat	Demographics and jobs	PublicEntity, ReportingPeriod, headcount, demographic breakdown, jobs created, job type
CommentThread	Feedback anchored to something	Related object (Target/TargetResult/DocumentRecord), author, body, parent, created date, resolved
TaskItem	Assigned work	PublicEntity, assigned to, assigned by, description, due date, status, internal/external
NotificationRule	Drives countdowns	ReportingPeriod, trigger offset (30d/15d/hourly), channel, recipient role

Notes on the model

The chain is the foreign keys. Allocation → Target → TargetResult →

DocumentRecord/ExtractionResult is what makes the accountability chain more than a slogan. Rand per outcome is a query across that path.

ExtractionResult is deliberately separate from TargetResult. Machine-read values live in their own table with confidence and source location until a human confirms them into the record. This separation is what lets us claim human-in-the-loop credibly rather than rhetorically.

RiskSignal is separate from RiskScore for the same reason: the score must be decomposable on screen into the signals that produced it.

Targets are versioned, not editable. Under section 4.4.4 of the 2019 Revised Framework for Strategic Plans and Annual Performance Plans, an in-year target may only change where the strategic plan is revised or an in-year budget adjustment occurs, approved by the executive authority and given effect by re-tabling the APP in the legislature. Being about to miss a target is not a lawful reason to change it.

So Target carries version, supersedes, effectiveFrom, supersededOn, revisionTrigger and retabblingReference, and the revision trigger enum has exactly two lawful values. A mutable counter would record that a target moved but not to what, when, or on whose authority, which is precisely the gap that makes performance unfalsifiable.

Audit outcome sits on its own record, not on a finding. An entity with a clean audit has no findings, so hanging the outcome off a finding row meant the two most important cases were unrepresentable: a clean audit, and an audit that was never completed at all. The outcome enum therefore carries OUTSTANDING alongside the five opinions, and it is the worst value in the set.

8. Technical architecture

Platform

Java 21, Spring Boot 3.3, PostgreSQL 16. Hibernate and Spring Data JPA for persistence with `ddl-auto: validate`, Flyway for migrations, Apache POI for Office document handling, Thymeleaf for the low-bandwidth surfaces, React for the office dashboard.

Firestore Authentication for identity only. It issues and verifies the token and holds no performance data. Two custom claims carry authorisation: `role` and `entityId`.

This is a deliberate change from the earlier draft of this document, which was written around Shesha, an ASP.NET Core low-code framework, on the understanding that using it was mandatory. It is not mandatory, and nobody on the team knows it. A framework the team cannot debug at hour 30 is a risk that dwarfs whatever it saves at hour 3.

Why the database matters more than the framework

Everything of value lives in PostgreSQL: the entities, the targets and their version history, the extraction rows, the confirmed results, the evidence, the risk scores and the signals behind them. Firestore is replaceable in an afternoon, because the only thing it produces is a verified identity and the security boundary is the claim on the token rather than anything Firestore stores.

That is what makes “the department can take this in-house” a true statement rather than a hedge. Ordinary Postgres, ordinary Flyway migrations, an ordinary Spring Boot jar. No proprietary runtime, no low-code designer to license, no vendor whose pricing changes as the entity count grows.

What we build

TemplateParser. Apache POI reads the reporting template and writes `ExtractionResult` rows, each carrying the cell it came from, such as `Quarterly Report!H14`. No performance data is written at this stage, and that separation is the human-in-the-loop guarantee expressed as schema rather than as policy.

SubmissionService. The upload, parse, confirm, submit, review lifecycle. There is no code path anywhere in the application that edits a confirmed figure. Not a disabled button, not a permission check. No endpoint.

RiskEngine. Deterministic and rule-weighted, and deliberately free of Spring and JPA so the arithmetic can be unit-tested without a database or a container. Seventeen tests pass. Two of them reproduce real published outcomes from real published facts.

UnitCostService. Planned against actual, entity against its own plan and its own history, peer comparison bounded to sector. It exposes no method for cross-sector comparison, and the absence is the design decision.

ExportService and ExportController. The filing in three shapes: full provenance JSON, a spreadsheet carrying source cells and evidence status per row, and a flattened DPME-shaped CSV. This is the piece that implements the position in §2.

NotificationService. The 30-day, 15-day and hourly countdowns, driven off the reporting period dates and their recorded legal basis.

PublicationService. The citizen projection, gated on `publiclyVisible` per entity, which ships false for every seeded body.

Low-bandwidth surfaces. Six server-rendered Thymeleaf pages, no JavaScript framework, every one under 5KB before compression.

Deadlines carry their authority

A late submission is a fact about an entity, so the system has to be right about what it was late for.

Treasury Regulation 26.1.1 gives 30 days after quarter end and covers actual and projected revenue and expenditure only. Treasury Regulation 30.2.1 requires quarterly performance reporting to the executive authority and sets **no day count at all**. The 30 days everyone quotes for performance comes from a National Treasury guideline, which binds as an instruction rather than as a regulation. The annual dates are the statutory ones: PFMA s55(1)(c) two months to the auditors, s55(1)(d) five months for the annual report, s65(1)(a) one month from receipt of the audit report, s65(2) a six-month backstop.

So every reporting period stores a `deadlineBasis` and a citation beside its dates, the field for a regulatory quarterly performance deadline is nullable because none exists, and the risk engine caps lateness against a departmental instruction at 60 percent of the signal while a missed statutory date can drive it to full. An entity chronically late on an instruction should never outrank one that breached the Act.

Microsoft technologies

Requirement (d) asks for integration with Microsoft technologies. Three layers, and honesty about which is which.

Built, and it is the spine of the product: Office is the interchange format. The reporting template is a real `.xlsx` workbook with three sheets and 51 formulas, generated by the system and parsed by it on the way back in. Government runs on Excel. Entities already compile these figures in Excel. Meeting them in the tool they use, rather than demanding they retype into a web form, is the integration that actually changes behaviour, and it is also what makes reliable parsing possible at all. Submissions export back to `.xlsx` carrying the source cell and evidence status on every row.

Configuration rather than code: Entra ID single sign-on. Firebase Authentication federates to Microsoft Entra ID as an OIDC provider, so DSAC staff can sign in with the government Microsoft accounts they already have. Our backend validates a token and reads two claims off it; it neither knows nor cares which provider issued it. That makes this a configuration change rather than a rebuild. The pitch should say exactly that, and should not imply it is live.

Roadmap, described as roadmap: SharePoint and Teams. Graph API document sync and Teams presence are not built. `DocumentRecord` holds a `storagePath` rather than file bytes, so a Graph-backed store is a driver swap behind a single interface. Say that much and no more.

Sovereignty position

The earlier draft rested this on the framework being South African. That argument left when Shesha did, and the replacement is stronger because it rests on the architecture rather than on a supplier's passport.

Standard Java, Spring Boot and PostgreSQL are among the most widely held skills in the South African developer market, so the department can hire maintainers rather than depend on one vendor. Cloud SQL for PostgreSQL is available in `africa-south1`, Johannesburg, so data residency is a deployment choice rather than a compromise, though the region is fixed at database creation and cannot be changed afterwards. The database is ordinary Postgres with ordinary migrations, so the system moves on-premise or to another provider without a rewrite. No per-seat licence as the entity count grows.

The one dependency worth naming out loud before a judge names it is Firebase. The answer is that it issues identity and nothing else. Every figure, every document and every audit trail sits in Postgres.

9. Security, privacy and responsible AI

Requirement (e) of the brief, and criterion 6 of the adjudication (10 percent).

Access control

Role-based, carried as Firebase custom claims and mapped to Spring Security authorities. Four roles:

- **Entity reporter.** Sees and submits only for their own entity.
- **DSAC reviewer.** Sees all entities, reviews and returns submissions, cannot alter submitted figures.
- **DSAC executive.** Read and summary access across the portfolio.
- **Administrator.** User and configuration management, no ability to alter submitted performance data.

The public surface is a separate unauthenticated projection exposing only a deliberately narrow, non-personal subset.

`entityId` lives on the signed token rather than in a request parameter, precisely so a client cannot choose it. A reporter reaching their own entity and nobody else's is a single comparison, routed through one helper that every controller uses, and it is the whole of the tenancy model. A caller who reaches another entity's record gets a not-found rather than a forbidden, because telling someone a record exists but is not theirs is itself a disclosure about another entity.

Data protection

POPIA-aligned, and scoped precisely rather than claimed in general, because a blanket compliance claim for a system that mostly does not process personal information is both wrong and easy to puncture.

POPIA applies to personal information. Performance data, being indicators, targets, actuals and narrative, is not personal information, so the eight conditions do not attach to the reporting payload at all. They attach to the identity and audit layer: official names, email addresses, and the confirmer and reviewer identities recorded on every row. Sections 19 on security safeguards and 22 on breach notification are the operative ones there.

The remaining personal information in scope is limited to reporter contact details and aggregated workforce demographics. Demographics are held and displayed in aggregate only, never at individual level.

Encryption in transit and at rest. Full audit trail on every change to a submission, target or document, with actor and timestamp. Data residency in South Africa, which the deployment model permits by default.

Responsible AI

Three commitments, each enforced by the architecture rather than by policy alone.

Nothing machine-derived enters the record unconfirmed. Extraction results live in a separate table with confidence and source location. A named human confirms them into TargetResult. The separation is structural, not procedural.

Every figure retains its provenance. Each extracted value points back to the document and location it came from. A reviewer can always open the source.

Risk scores are decomposable and deterministic. The score is rule-weighted, not a trained black box. The interface always shows the contributing signals. A reviewer can disagree with any one of them, and the reasoning is auditable after the fact.

Deliberate limits

The system does not auto-approve submissions, does not publish anything to the public surface that has not been through DSAC review, and does not generate performance judgements presented as fact. It surfaces signals; people decide.

10. Judging criteria mapping

#	Criterion	Weight	What earns it
1	Relevance to the challenge	20%	All five mandatory requirements built (\$4). Partial implementation is explicitly allowed where the approach is credible, so anything descoped is presented with a concrete path.

#	Criterion	Weight	What earns it
2	Innovation and emerging technologies	15%	A machine-readable reporting standard where none exists, and an evidence layer under it. Every figure carries its source cell and its named confirmer, which is the Auditor-General's reliability test built as schema. Deterministic explainable scoring that reproduces published audit outcomes. Positioned to feed DPME's eQPRS rather than duplicate it.
3	Technical feasibility and functionality	20%	A working end-to-end path, not slides. Submission actually flows to dashboard to citizen view in the demo. Real published allocations for all 28 entities, reconciling exactly to Treasury's totals in four consecutive years. Integration shown through the export layer and the Excel round trip.
4	UX, accessibility, inclusivity	10%	Sub-100KB mobile submission view for under-resourced NPOs, low-bandwidth citizen page, plain language, offline-tolerant drafts. A working channel, not a font-size claim.
5	Data, intelligence, insight	10%	Unit cost, planned versus actual. Risk scores shown with their contributing signals. Every figure traceable to a source cell. Transparency is structural, not asserted.

#	Criterion	Weight	What earns it
6	Security, governance, responsible tech	10%	Role-based access, POPIA alignment, audit trails, human-in-the-loop enforced by the data model, deterministic explainable scoring, publication gated behind departmental approval.
7	Scalability, sustainability, sovereignty	10%	Standard Java, Spring Boot and Postgres, which are the commonest skills in the local market. Data resident in Johannesburg. Everything of value in an ordinary database the department can take in-house, with identity the only external dependency. No per-seat licensing. The standard itself is reusable by any department.
8	Presentation and demonstration	5%	The 90-second thread in §6.5, rehearsed three times and recorded as backup. One story, three audiences, no slide needed.

Where the points actually are

Only 20 percent is tied to the specific challenge statement. Another 45 percent — innovation, data and insight, security, sovereignty — rewards how we build and how we explain it, and is available to any team that thinks about it. Most will not.

The implication for the build: get the mandatory floor working first, because criteria 1 and 3 together are 40 percent and both require things that visibly function. Then spend remaining time on explainability and the public view, which are cheap to build and carry disproportionate weight.

11. The 48-hour build

Hours run from T0, the moment the challenge is assigned. Assumes a team of five. Every block ends at a point where something is demonstrable, so that a block overrunning costs depth rather than the whole submission.

The governing rule. Criteria 1 and 3 are 40 percent combined and both require things that visibly work. Day one ends with all five mandatory requirements present, even if shallow. Nothing in day two starts until day one's list is green.

Roles, assigned at T0 and not rotated

Role	Owns
Domain	Entity model, migrations, generated APIs, seed data. Critical path for the first eight hours.
Config	Form-builder pages, dashboards, roles, repository, workspaces. Most of the mandatory floor.
Custom	React components: risk panel, unit cost view, citizen page.
Services	Template parser, risk scoring job, notification scheduler, mobile submission view.
Pitch	Real data sourcing first, then narrative, demo script and rehearsals. A full role, not a spare pair of hands.

H0-H3 · Spike and split

Three things happen at once.

1. Get it serving (Domain). The provided project already contains the entities, the migrations and the real seed data. Run `docker compose up -d` and `mvn spring-boot:run`, then open the citizen view, which needs no authentication and is therefore the fastest proof the application is alive.

This is a go/no-go gate. The project was written in an environment that could not reach Maven Central and has never been compiled end to end, so expect the first build to surface a handful of import and signature fixes. If it is not serving by H3, stop adding anything and spend the time getting a clean compile.

2. Real data (Pitch + Custom). Pull actual figures from published DSAC entity annual reports and APPs — four to six entities, three years of targets, allocations and Auditor-General outcomes. These are public documents. **This is the highest-leverage three hours in the build.** A risk score computed over invented history is circular and a judge can say so in one sentence; computed over real history it is a finding.

3. The template (Services + Config). Draft the reporting template: one row per indicator, columns matching the schema. This artifact drives the data model, the parser and the demo, so it must exist before the model is frozen.

Exit: hello-world running, real data in a spreadsheet, template v1 agreed.

H3-H8 · Domain and data

- **Domain:** entity classes from §7 implemented, migrations run, generated CRUD APIs on, four roles configured.
- **Pitch + Custom:** finish loading real data as seed.

- **Services:** template parser written against template v1 while the model settles.
- **Config:** repository and entity list pages started.

Exit at H8: every entity in §7 exists, APIs respond, real data is queryable, roles are enforced.

H8-H16 · The mandatory floor

All five of the brief’s requirements become demonstrably present. Shallow is acceptable; absent is not.

Requirement	Owner	Done means
(c) Document repository	Config	Upload, version on re-upload, approval state, per entity
(a) Analytics	Config	Per-entity dashboard: targets by state, year-on-year, audit findings, workforce stats
(b) Early warning	Services	Countdowns firing at 30 days, 15 days, hourly, off real due dates
(d) Workspaces	Config	Per-entity space, comments anchored to targets, task assignment
(e) Security	Domain	Four roles enforced, audit trail on every change
Submission flow	Custom + Services	Upload template, parse, confirm, submit

Two cheap wins to take deliberately. Task assignment is named in requirement (d) and is roughly thirty minutes of CRUD in this framework — build it, it is free relevance. And “comments visible in real time” is satisfied by a five-second poll; do not build a websocket layer for a demo.

Exit at H16 — the day-one gate. Walk the entire floor end to end, out loud, in front of each other. Anything missing gets fixed before anyone sleeps. Everything after this point is optional by comparison.

H16-H22 · Sleep

Rotate one person on a light shift if you want cover. Everyone else sleeps. A team that skips this writes the bugs it will spend H30 to H40 hunting. This block is part of the plan, not a failure of it.

H22-H30 · Intelligence

- **Services:** risk scoring job. Compute RiskSignals from the real history, weight them, write RiskScore. Deterministic and rule-based — you must be able to explain any score on a whiteboard without opening the code.
- **Custom:** the risk panel. Score with its contributing signals expanded beneath it. This one component carries criterion 5 and half of criterion 6, and deserves disproportionate effort.

- **Domain:** unit cost query across Allocation → Target → TargetResult.
- **Config:** portfolio view with entities ranked by risk rather than alphabetically.
- **Pitch:** first draft of the demo script, rehearsed against whatever exists.

Exit at H30: open an entity, see a risk score, expand it, understand why.

H30-H36 · The differentiators

- **Custom:** unit cost view — planned versus actual, entity against its own plan and its own history, peer comparison only within sector grouping.
- **Custom:** citizen page. Unauthenticated, minimal payload, publication gated behind a DSAC flag.
- **Services:** low-bandwidth mobile submission view.
- **Built:** Excel export of a submission. Two sheets: the filing with the source cell and evidence status on every row, and a provenance sheet carrying the deadline authority and the caveats. This is the substantive Microsoft-technologies claim and the report-once story in its simplest form, and it closes the loop that the `.xlsx` template opens.
- **Built:** DPME-shaped export, so the same filing goes to eQPRS without retyping.

Exit at H36: the full demo thread runs start to finish, even if rough.

H36-H40 · The flourish, conditional

Two candidates, in this order, **if and only if** H36's exit was clean.

Entra ID sign-in. Configure Firebase Authentication to federate to Microsoft Entra ID and sign in once with a Microsoft account on stage. This is a console configuration rather than code, and it converts the middle layer of the Microsoft answer from described to demonstrated. Timebox to 90 minutes; if the tenant is not cooperating, drop it and describe it instead, which is still an honest answer.

WhatsApp over a sandbox number. Lower priority than it was, because the mobile view already carries the inclusivity claim and carries it with measured page weights. If the webhook is not receiving by H39, drop it and say in the pitch that the channel is next.

Otherwise spend this block polishing the demo path and fixing what breaks under repetition.

H40 · Feature freeze

Hard stop. No new functionality after this line, whatever state it is in. The most common way a good hackathon project loses is a half-finished feature breaking a working demo at hour 46.

H40-H44 · Rehearse and insure

1. **Record the demo end to end on video.** Conference wifi fails, laptops sleep, databases lock. A recording is the insurance policy and costs twenty minutes.
2. **Rehearse three times, out loud, on the clock.** Not a read-through — the full ninety seconds plus the questions after.

3. **Fix only what breaks on the demo path.** A bug on a screen the judges will not see is not a bug this weekend.
4. **Prepare the four answers you will certainly be asked:**
 - *Where does the data come from?* Real published annual reports, named.
 - *Does the AI decide anything?* No. Parsed values sit in a separate table until a named human confirms them.
 - *Is this a new reporting burden?* No. The template digitises what entities already produce.
 - *Why should the state trust it long-term?* Standard Java, Spring Boot and Postgres, data resident in Johannesburg, everything of value in an ordinary database, no per-seat licence, the department can take it in-house.
 - *How is this different from eQPRS?* eQPRS captures numbers and does not hold the evidence behind them, and it was built around departments while every DSAC body is Schedule 3A, reporting to its executive authority under a regulation that sets no deadline and names no system. We fill that gap and export into theirs.
 - *Where is the Microsoft integration?* The reporting template is a real Excel workbook the system generates and parses, and submissions export back to Excel with the source cell on every row. Entra ID sign-in is a Firebase federation setting rather than a rebuild. SharePoint and Teams are roadmap, and we say so.

H44-H48 · Buffer and present

Sleep if the schedule allows. Arrive early. Test the room's projector and network before you need them. Keep the video on a local drive, not behind a link.

The cut list, in the order things go

1. WhatsApp channel → mobile view ships, channel becomes roadmap
2. Entra ID live sign-in → described rather than demonstrated
3. Peer comparison → entity against its own plan only
4. Citizen page → one entity rather than all of them
5. Executive narrative summary → cut entirely, invisible in ninety seconds

Never cut: the end-to-end demo thread, the risk panel's visible reasoning, the real seed data, or the Excel round trip. Those four are the submission. The Excel round trip is on this list and not on the one above because it is the only thing standing behind requirement (d).

12. Stress test

The nine questions worth asking about this plan before a judge asks them, and the answers. Five of these changed the plan; the rest are Q&A preparation.

Is comparing cost per outcome across entities defensible? No, and the original version of this document was wrong to propose it. A ballet company and a sports federation do not produce commensurable outputs, and anyone with a public finance background will say so within thirty

seconds. Fixed in §5.1: the primary comparison is an entity against its own plan, which is its own commitment and therefore unarguable. Peer comparison survives only within sector groupings and carries a label saying it is indicative.

Can arbitrary document extraction be built and demoed reliably in 48 hours? No. Fixed in §5.3 by defining a template and parsing that. This turned out to improve the product rather than compromise it — the missing thing in the real world is the standard, not the portal — and it is now the spine of the concept in §2.

Will a public transparency layer be read as naive? Yes, if we publish by default. Publishing entity underperformance is a ministerial decision, not a product decision. Fixed in §5.6: the view is publish-ready and gated behind a departmental flag. Understanding that distinction is itself a signal that we know how government works.

Is a WhatsApp channel achievable? Not in production form, because Meta business verification takes weeks. A sandbox number works for a demo. Fixed in §5.5 by shipping a low-bandwidth mobile view first, which now carries the inclusivity claim on its own and carries it with measured page weights rather than a promise. The channel is a conditional addition and the constraint is stated in the pitch rather than glossed.

Will a risk score over invented data survive scrutiny? No. Scoring patterns we fabricated is circular and a judge can say so in one sentence. This is now resolved rather than planned: all 28 entities carry their real transfer allocations from Estimates of National Expenditure 2026, Vote 37, Table 37.3, which reconcile exactly to the published portfolio totals in all four years, and sixteen carry published audit outcomes. The engine reproduces Robben Island Museum as the worst entity in the portfolio and SAHRA as among the best, from published facts alone, without being told the answer. That is the demonstration.

The one thing still illustrative is quarterly submission timing, because per-entity quarterly performance is not published at this granularity. It is labelled as such in the code and should be volunteered in the pitch rather than discovered.

Can the team be productive in an unfamiliar framework? This was the largest schedule risk in the plan while the build was pinned to Shesha. It was resolved by establishing that Shesha is not mandatory and moving to Java and Spring Boot, which the team knows. The go/no-go gate at H3 in §11 remains, now aimed at getting the provided project to serve rather than at learning a framework under time pressure.

Are we duplicating something government already runs? Partly, and pretending otherwise is the fastest way to lose the room. DPME's eQPRS already captures quarterly performance against targets and draws dashboards. Fixed in §2: we do not claim capture as the innovation. The claim is the evidence layer underneath it, the Schedule 3A reporting line that no system serves, and an export that feeds eQPRS instead of competing with it.

Where is the Microsoft integration, really? One layer built, one a configuration change, one roadmap, set out in §8. The honest summary is that Office document handling is not a token gesture here, it is the spine: the template is a real Excel workbook, entities already work in Excel, and meeting them there is what makes both adoption and reliable parsing possible. If a judge wants a Teams tab, we do not have one and we say so.

Is the lateness signal measuring something real? It was not, in the first version. It scored entities against a 30-day quarterly performance deadline that does not exist in regulation; the 30 days in Treasury Regulation 26.1.1 covers revenue and expenditure only. Fixed in §8: deadlines now store their legal basis, lateness against a departmental instruction is capped, and only a missed statutory PFMA date can drive the signal to full.

What is the single point of failure? Anything live at a conference venue. Handled: everything runs locally, and a full demo recording is made at H40 before the freeze.

Are we over-claiming? The earlier draft repeatedly asserted that no other team would build a given feature. We do not know that, and saying it in the room is a bad look. The pitch states what we built and why it matters, and lets the judges do the comparing.

Are we adding work for entities already drowning in reporting? A fair challenge and the answer must be ready. The template digitises the performance information entities already compile; it does not introduce a sixth report. Stated in §2 and in the prepared answers in §11.

13. Open questions and assumptions

Answer before T0

Team size and skill split. How many are comfortable in Java and Spring Boot, how many in React, and who owns the demo. The role assignments in §11 depend on this.

Does the build machine have Maven access? The project has never been compiled end to end because the environment it was written in could not reach Maven Central. Confirm someone can run `mvn spring-boot:run` before the clock starts, because that is the H3 gate and there is no way around it.

Can anyone confirm the eQPRS export columns? Ours is a reading of a published reporting shape, not a certified integration. If a DPME contact is reachable, this is worth one question.

Judging format and time limit. The demo script in §6.5 assumes a live walkthrough plus questions. Confirm.

Assumptions made in this document

- No live integration to any departmental or Treasury system; everything is seeded or uploaded
- Microsoft integration is graded rather than a pass/fail gate, and partial implementation with a credible path scores
- “Full functionality on mobile devices” in requirement (d) is satisfied by responsive web
- The reporting template is derived from the APP structure entities already use. **Verify the exact name and current version of the governing performance-information framework before quoting it in the pitch** — claiming alignment with a document whose title is wrong is worse than not citing one.

Deliberately out of scope

- Financial system integration or transaction-level expenditure tracking
- Any workflow that certifies or signs off performance information for audit purposes
- Multi-department extension beyond DSAC, though the model supports it and that is worth saying aloud